Does Independent Retrieval Help an LLM Fact-Checker?
A Staged Evidence-First Validator for Scientific Claims

Abstract

A checker that endorses a scientific claim the evidence does not support is worse than a checker that abstains, so verification systems are often designed to gather their own evidence rather than trust the citations they are handed. We build such a system and measure it. Our validator stages four isolated roles over two evidence scopes: a sealed reader that answers a neutral question from retrieved passages before seeing the claim, a claim-visible judge over four labels (supported, contradicted, unaddressed, underdetermined), a citation auditor over the original references, and a reconciler that reports the two scopes separately instead of collapsing them into one score. Every stage fails closed: a transport or parse failure is an execution status, never a scientific verdict.

On SciFact, with a frozen corpus, locked splits, and a preregistered primary comparison, we find *no significant reduction in false endorsement* from independent retrieval plus staged judgment over a same-evidence baseline that already reads the original citations. On the 300-claim development sample the paired difference in false endorsement is -0.012 ([-0.031, 0.000]), and the same pattern repeats on a 300-claim held-out sample (-0.013, [-0.033, 0.000]). The staged validator does eliminate false endorsement, but it does so by endorsing much less: SUPPORT recall falls from 0.530 to 0.383 while 32 of 268 scored claims fail closed. We replicate the pattern with a second, architecturally different model family, and we show that the dominant failure mode is not retrieval but a systematic disagreement between the two evidence scopes. We report this as a negative result with a mechanism analysis, a cost accounting, and a fully reproducible artifact. The design lesson is narrow and useful: independent retrieval is not sufficient for scientific claim verification, and evidence-first staging can convert a noisy checker into a conservative one without converting it into a correct one.

Artifact. All tables and figures in this paper are generated from saved prediction files by the scripts in `scripts/paper/`. The corpus, splits, prompts, and frozen configurations are in the repository (`github.com/`), and the held-out split is protected by a hash freeze that refuses a run whose model, prompt, configuration, or corpus differs from the locked record.

1 Introduction

Large language models are now asked to check claims, not only to make them. The usual pipeline is simple: a checker sees a claim, sees the citations that were offered with it, and returns a verdict. This design has a structural weakness. The checker inherits the generator's evidence: if the generator cited the wrong paper, or cited a paper that supports a slightly different claim, the checker is reading the same biased evidence set and is structurally unable to notice. The failure is not a bug in the checker; it is a property of the information it is given.

The obvious remedy is to make the checker search for its own evidence. Recent work in retrieval-augmented factuality evaluation [**?**], in self-check agents [**??**], and in retrieval-augmented verifiers [**??**] all point in that direction. It is tempting to assume that independent retrieval plus careful staging must help. That assumption has rarely been tested against a strong same-evidence baseline on scientific claims, where the original citations are often correct and where a well-calibrated abstention is genuinely valuable.

This paper tests the assumption directly. We build a staged validator (**V**) that separates four roles – a sealed evidence reader, a claim-visible judge, a citation auditor, and a reconciler – over two evidence scopes: the original citations (D_0) and independently retrieved abstracts (D_1). We compare it against a

same-evidence baseline (**B2**) that reads a neutral question over the original citations in an isolated reader and then compares, which is the strongest same-evidence configuration we could build without training. We also build the full factorial around the two interventions we care about: evidence access (D_0 versus D_1) and reading policy (claim-visible versus neutral-question evidence-first).

Our preregistered primary outcome is false endorsement: the rate at which a method predicts "supported" for a claim that the gold labels do not support. The answer is *no*. On the frozen development sample the paired difference between V and B2 is -0.012 with a 95% interval of [-0.031, 0.000], which includes zero. The held-out sample repeats the pattern. The staged validator does reach zero false endorsements, but it buys that by withholding support: SUPPORT recall drops from 0.530 to 0.383 and 32 claims fail closed entirely. A checker that refuses to answer is not a better checker.

Three findings follow, and we think they are more useful than a headline number.

- **Independent retrieval is not the bottleneck; scope reconciliation is.** The failure analysis shows that the dominant mode of V is not a retrieval miss but a disagreement between the D0 and D1 judgments, which the reconciler resolves conservatively into "underdetermined". When the two scopes agree, V inherits the baseline's accuracy; when they disagree, V abstains. The mechanism is a disagreement detector, not an evidence-acquisition improvement.
- **Staged judgment buys precision, not accuracy.** Staging changes what the system is willing to say, not how often it is right. Reported as F1 alone, this looks like a regression (0.747 against 0.675); reported as a precision–coverage trade-off it is a usable operating point for a conservative reader-facing tool.
- **The result is not model-specific.** We reproduce the pattern with a second model family accessed through a different harness. This matters because the negative result would be weak evidence about verification design if it were a single model's idiosyncrasy.

We also state what we do *not* do. We do not report a human study: no participant was recruited, no ethics review was obtained, and no claim about human trust is made. We do not report four-way gold labels: SciFact's label space is three-way, so our "unaddressed" and "underdetermined" distinction is described but not scored against human annotation. We do not claim reproduction of trained verifiers; the baselines here are named prompt-level adaptations, and we say so each time. Finally, our corpus is a fixed set of biomedical abstracts, so every claim of support is relative to that corpus and that retrieval budget.

The contribution of this paper is therefore a negative result with a mechanism, not a new score: a reproducible measurement of what independent retrieval and evidence-first staging do, and do not do, to a scientific claim checker, together with the artifact that makes the measurement checkable.

2 Related Work

2.1 Verification datasets and scientific claim checking

FEVER [**?**] established the claim–evidence verification format that this paper follows, and FEVEROUS [**?**] extended it to unstructured and structured evidence. SciFact [**?**] is the benchmark used here: it pairs scientific claims with abstracts and rationale sentences, and supplies support, refute, and insufficient-information labels. SciFact-Open [**?**] extends the task to open-domain retrieval, which motivates our separation of a frozen corpus from live search; we use a frozen corpus only, and we do not claim open-domain performance. Missci [**?**] targets misrepresented science and fallacy reconstruction rather than four-way claim labeling, and RAGTruth [**?**] targets context faithfulness of retrieval-augmented generation rather than evidence sufficiency of a checker; we cite both as boundaries of our setting rather than as benchmarks we score. For biomedical claims specifically, SciClaims [**?**] provides an end-to-end generative pipeline; we treat it as prior art for scientific verification with explanations and do not reproduce it. Vladika and Matthes [**?**] compare knowledge sources for open-domain scientific claim verification and motivate our interest in whether a different evidence source changes verdicts.

2.2 LLM verifiers, self-checking agents, and retrieval

A long line of work asks an LLM to judge claims. Chain-of-Verification [**?**] isolates the verification questions from the draft answer, which is the closest published analogue to our information separation and which our B2 baseline deliberately imitates. Safe [**?**] verifies atomic facts with search and is the reference point for our fresh retrieval arm. Self-checking agents such as MARCH [**??**] add information separation between proposer and checker, and blame-localization work [**?**] shows that trace-level attribution of citation errors is a developer-facing diagnostic distinct from evaluating a final conclusion for an end user; we evaluate final conclusions, and cite the trace work as the reason not to claim it. A systematic auditor for retrieval-augmented generation [**?**] and retrieval-augmented verifiers such as RAFTS [**?**] and RAV [**?**] further motivate separating evidence discovery from judgment. Explainable fact checking through question answering [**?**] and hallucination detection methods such as SelfCheckGPT [**?**] and RARR [**?**] address the neighbouring problem of detecting unsupported content in model output.

We are explicit about what is *not* new here. Atomic decomposition, blinded checking, search-based verification, and readable explanations each have substantial prior art. Our contribution is a controlled measurement of their combination against a strong same-evidence control, and a negative result about that combination.

2.3 Abstention, calibration, and selective use

A system that cannot say "I do not know" is dangerous. Work on abstention [**?**] and on calibrated selective fact checking [**?**] formalises the trade-off we measure. Calibration theory [**?**] distinguishes a model's verbalised confidence from a probability fitted on held-out data; we fit a single temperature on a calibration split and report raw and calibrated numbers separately. Because a scientific "underdetermined" verdict and a system-level "defer to a reviewer" action are different things, we keep them in separate fields. On the human side, confidence and explanation are known to affect both accuracy and trust calibration in AI-assisted decisions [**?**], so we do not claim that human trust calibration is unstudied; we simply did not run the study and we report the protocol without results.

3 Problem Formulation

3.1 Inputs and evidence scopes

Let c be an atomic scientific claim taken from a frozen report. A claim carries an exact source span, a normalised statement, and a *neutral question*: a single interrogative formulation of the claim that preserves population, comparator, units, and causal scope but does not contain the asserted answer. Two evidence scopes are available:

- D_0 — the documents the original report cited. These are the generator's evidence. A judgment over D_0 answers: *is the original citation set faithful to the claim?*
- D_1 — documents retrieved independently from a frozen corpus using the neutral question, with the claim's answer, the report, and D_0 hidden from the retrieval and reading stages. A judgment over D_1 answers: *what does independent retrieval establish?*

A verdict is always relative to a stated scope, corpus, and search budget. It is not an unrestricted declaration of scientific truth. The reconciled judgment over $D_0 \cup D_1$ answers: *what can be said once both scopes are considered?*

3.2 Four labels

Internally the system uses four labels, defined relative to a specified evidence collection:

Supported
Relevant evidence establishes the exact scoped assertion and no unresolved material counterevidence remains in the reviewed pool.

Contradicted
Relevant evidence establishes an incompatible assertion; apparent conflicts have been resolved by scope or relevance.

Unaddressed
No reviewed relevant passage answers the claim under the stated corpus and retrieval budget.

Underdetermined
Relevant evidence exists, but disagreement, imprecision, incomplete measurement, or an unsupported inference prevents resolution.

The decision order is: check execution status and claim validity; then assess relevance; if no passage addresses the claim, use *unaddressed*; if relevant evidence is materially conflicting or inconclusive, use *underdetermined*; otherwise apply support or contradiction, but only at the preserved scope. A non-significant result is not automatically proof of no effect, and a null-result claim requires its own evidence assessment.

SciFact's native label space is three-way, so the scored mapping is supported→SUPPORT, contradicted→REFUTE, and unaddressed/underdetermined→NEI. The four-way distinction is therefore *reported but not scored* against gold. We say this once here and repeat it wherever a number could be misread.

3.3 Research questions

- **RQ1 (primary, preregistered).** At a matched inference budget, does independent retrieval plus staged evidence-first judgment reduce false endorsement relative to same-evidence checking?
- **RQ2 (descriptive).** Can the system distinguish missing evidence from unresolved evidence, and how often does it defer?
- **RQ3 (deferred).** Do evidence-linked explanations improve non-experts' judgments, including when the validator is wrong? No participant was recruited; Appendix D gives the protocol and no results.
- **RQ4 (secondary).** How do accuracy, coverage, latency, and cost change across generator/checker families and across the two interventions?

4 System Design

Figure 1 shows the staged pipeline and its information boundaries. Roles are isolated functions with typed inputs; several roles may call the same underlying model in separate stateless requests. Separate roles do not imply statistically independent errors, which is exactly why model-family dependence is measured rather than assumed.

4.1 Roles and information boundaries

Proposer.
Maps a report to atomic claims, neutral questions, asserted answers, and exact source spans. Quantities, confidence language, population, comparator, and causal scope are preserved; a conjunction is split while the relation between its parts is retained.

Independent gatherer.
Given the neutral question, a scope, and the corpus, retrieves fresh evidence. The asserted answer, the report, the original citations, and the generator's identity are hidden. Three query forms are issued

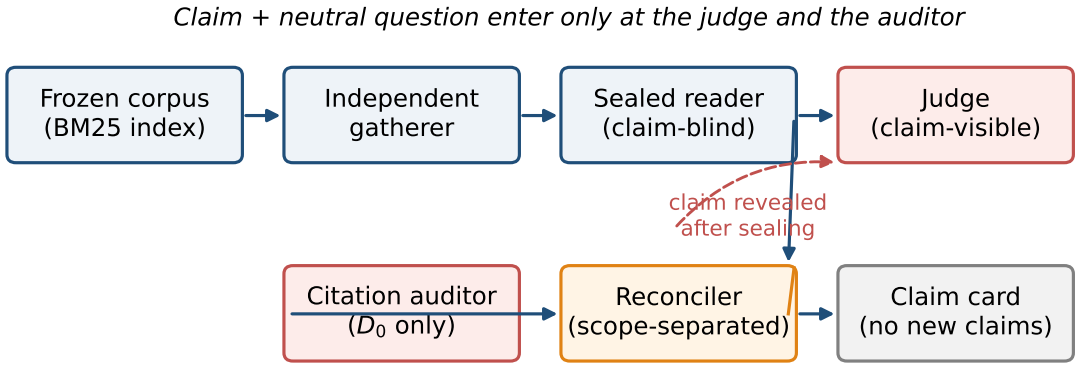


Figure 1: The staged validator. Solid arrows carry evidence; the dashed arrow is the only path by which the claim reaches the judge, and the reader's sealed record is fixed before the claim is revealed. D0 (original citations) and D1 (independent retrieval) are judged separately; the reconciler reports both scopes and never repairs one with the other. Any transport or parse failure yields a failed execution status, not a scientific label.

Table 1: The validation procedure for one claim. "Scope" is the evidence collection a step may read; no step may read outside its scope. A failure in any step produces an execution status and stops the claim.

Step	Role	Reads	Writes
1	Proposer	report	atomic claims, neutral questions
2	Gatherer	neutral question, corpus	D_1 passages
3	Reader	neutral question, D_1	sealed evidence record
4	Judge	claim, sealed record, D_1	four-way label, spans
5	Auditor	claim, D_0	D_0 label, citation adequacy
6	Reconciler	steps 4 and 5	reconciled label, conflicts
7	Renderer	reconciled record	claim card

per claim (open inquiry; scope and measurement; limitations or null results); the top candidates are deduplicated and reranked, and up to eight passages are kept.

Evidence reader.
Answers the neutral question from the shuffled retrieved passages and cites the passage identifiers it used. Its record is *sealed* before the claim is revealed.

Judge.
Sees the atomic claim, the sealed reader record, and the evidence passages, and assigns one of the four labels together with rationale codes, cited spans, and uncertainty reasons.

Citation auditor.
Judges the claim against D_0 alone and rates each citation's adequacy. It never sees D_1.

Reconciler.
Combines the two judgments, explains the difference, and flags invalid inference links between claims.

Renderer.
Produces plain-language claim cards from the validated structured record only. It introduces no new

factual claims.

The blinding is partial by construction and we describe it as such: a neutral question still reveals the topic and may leak assumptions. The isolation guarantee we test is structural – the reader's payload contains only the neutral question and passage text, enforced by an assertion on the payload shape and a test suite – not semantic.

4.2 Execution and stopping

The report is generated once and frozen. Validation does not rewrite it. A claim budget of eight claims per report, two retrieval rounds, three query forms per round, and eight passages in the judge context are development defaults that were frozen before the held-out run. A second retrieval round is permitted only for unresolved questions, scope mismatch, or contradictory evidence; a run stops on exhausted budget, no new eligible documents, or a complete bounded evidence record. Every intermediate artifact is persisted, and the runner resumes at the last completed claim without mixing configurations.

Fail-closed behaviour is the load-bearing design decision. When the model returns malformed JSON, cites a passage it was not given, or the endpoint fails, the system records an execution status of `failed` or `timeout` and a placeholder label that is a schema token, never model text. These rows are excluded from scored metrics and reported separately. They are never silently remapped into a scientific verdict.

4.3 Reconciliation

Agreement between scopes copies the shared label. A disagreement, or a failed D0 audit, resolves to *underdetermined* and hence to NEI on the scored label space. We deliberately do *not* repair citation faithfulness with outside evidence: a claim can be unsupported by its own citation and supported elsewhere, and the two facts are reported separately.

5 Implementation

5.1 Records and reproducibility contract

The implementation emits a machine-readable prediction file and an immutable run sidecar for every condition. The run record carries a run identifier, split, seed, model identifier and revision, prompt-bundle hash, corpus hash, configuration hash, timestamps, token totals, and an execution status; the sidecar adds the prompt files and their individual hashes. Predictions carry the claim identifier, the predicted label, the four-way label where applicable, the evidence scope, the rationale codes, and the execution status. Schema validation is enforced before a file is accepted.

The corpus is frozen by a SHA-256 hash over the canonicalised abstract corpus, and the split manifests are locked separately. The held-out split is protected by a freeze record containing the model identifier, the corpus hash, and per-method configuration and prompt hashes; a held-out run refuses to start unless the observed hashes match the frozen ones. This is what makes "we ran the held-out split once" a checkable statement rather than an assurance.

5.2 Retrieval

Retrieval is BM25 (Okapi, $k_1 = 1.5$, $b = 0.75$, epsilon idf floor) implemented in pure Python over the frozen abstract corpus, with a fixed alphanumeric lower-case tokenizer and no stemming. For each neutral question three query templates are used; the top 20 candidates per query are merged, deduplicated, and reranked by maximum BM25 score, and at most eight passages are passed to the judge. Adjacent sentences of one document are kept together where needed to resolve negation or units. Dense retrieval and reciprocal-rank fusion were prepared but are not enabled in the product path; they are development-time comparisons, and we do not report a dense retrieval result we did not measure.

5.3 Models and harness

We evaluate two model families to satisfy RQ4 and to protect the negative result from being an artefact of one model's behaviour.

GPT-6-LUNA (API).
A hosted instruction-tuned model accessed through an OpenAI-compatible chat endpoint.

MINIMAX M3.1 FLASH.
A second, architecturally unrelated instruction-tuned model, accessed through the OpenCode agent runtime. To keep the repository's endpoint policy intact we expose that runtime through a small loopback bridge that implements the OpenAI chat completions interface, one fresh session per request. A fresh session per request matters: a shared session would carry earlier claims into later ones and break the information-isolation guarantee the validator is measured on. The harness adds a fixed agent preamble and tool definitions to every request; the preamble is byte-identical across conditions, so it cancels in every comparison, but it does inflate the reported prompt-token counts, and we note this where we discuss cost.

Both families run with temperature 0, the same prompts, the same frozen corpus, the same claim sample, and the same budgets. The prompts are versioned text files under `prompts/`; their hashes enter the freeze record.

5.4 Failure handling and the retry policy

Three different things can go wrong, and the paper keeps them apart.

1. **Transport failure.** The endpoint refuses the connection or returns a retryable status. The client retries with exponential backoff.
2. **Protocol failure.** The model returns text that is not the requested JSON, cites a passage it was not given, or returns a label outside the allowed set. The parser rejects it. This is a *parse* failure, not a scientific disagreement.
3. **Scope conflict.** Both stages completed and disagree. This is a scientific outcome: the reconciled verdict is *underdetermined*, which maps to NEI.

The first two produce an execution status of `failed` and are excluded from the scored metrics; the third is scored normally. Because a failed request is not evidence about the claim, the runner retries it, up to four attempts per claim, and drops the failed row from the prediction file between attempts so that the resume re-claims exactly those claims. The policy is applied identically to every condition and every model family on the development sample.

Because dropping failed rows could flatter a method that fails on the hard claims, we report a second, stricter accounting in which every submitted row is scored with the label the method recorded for it. On the fail-closed paths that label is the conservative NEI, so the strict accounting can only hurt a method. Table 5 shows that the primary comparison does not depend on the choice.

One asymmetry is stated rather than hidden: the held-out run is the single execution it was always meant to be and does not use the retry policy, so its fail-closed count includes unrecovered execution failures.

5.5 Prompt contracts

Each role has a fixed output contract, expressed in the prompt and enforced by the parser. The reader must return an answer and the identifiers of the passages it used; the judge must return one of four labels, rationale codes, cited span identifiers, and uncertainty reasons; the citation auditor must return a label plus a per-citation adequacy rating from {adequate, inadequate, unverifiable} and a defect list. The parsers reject a label outside the allowed set, a cited identifier that was not supplied, and any missing field. Appendix A lists the prompts in full.

Table 2: The condition matrix. "Adaptation" means a prompt-level implementation of the named mechanism, never a reproduction of a trained system. The 2×2 factorial is B1/B2 (evidence D_0) against B3/V-D1 (evidence D_1), crossed with claim-visible against neutral-question evidence-first reading.

ID	Condition	Evidence and reading policy
B0	Prior-knowledge judge	No evidence; claim only
B1	Atomic claim, claim-visible judge	D_0; claim-visible
B2	Neutral question, isolated reader, compare	D_0; evidence-first
B3	Fresh retrieval, direct judge	D_1; claim-visible
V-D1	Fresh retrieval, isolated reader, judge	D_1; evidence-first
V	Full staged validator	$D_0 \cup D_1$; reconciled
O	Evidence oracle (diagnostic)	Gold rationale sentences

6 Experimental Setup

6.1 Data and splits

We use SciFact [**?**]. The corpus is the released set of biomedical abstracts, frozen by hash. The released training claims are split into a calibration subset of 100 claims, reserved before any prompt work, and a development pool of 709 claims. The public development split (300 claims) is used as a *local* held-out evaluation set; it is not an untouched community test set, and we name it accordingly throughout. Official test labels are not used.

Two samples are frozen by manifest: a 300-claim development sample (seeded draw, `data/manifests/dev300_s` used for all development and prompt decisions, and a 300-claim held-out sample used once under the freeze record. Both samples group claims by cited-document family so that paraphrases and derivative claims do not straddle the comparison.

6.2 Conditions

All conditions share the frozen reports, the corpus snapshot, the claim sample, the scoring code, and the prompt files. They differ only in the intervention named in the table.

B0 measures how much the model answers from parametric knowledge alone and acts as a floor. O supplies the gold rationale sentences to the same claim-visible judge and is a diagnostic ceiling; it is not a deployable competitor and we never describe it as one. V-D1 is the D1 arm of V, recovered from the V predictions by a mapping step rather than a second model run, so it inherits V's failure modes exactly.

6.3 Metrics

Verification quality is measured by per-class precision, recall and F1, plus micro and macro F1 on the native three-way label space. The primary metric is *false endorsement*: the number of claims predicted SUPPORT whose gold label is not support, divided by the number of gold non-support claims. We report the second denominator (false endorsements over all predicted SUPPORT claims) as well, because the two answer different questions. Coverage is the fraction of submitted claims that produced a scored prediction; fail-closed rows are counted and reported, never dropped silently. Calibration is measured with multiclass Brier score, negative log-likelihood, and expected calibration error, fitted with a single temperature on the calibration split only. Cost is reported as tokens, calls, and latency per claim.

6.4 Statistics and preregistration

One primary comparison was preregistered before the held-out run: V versus B2 on false endorsement with the gold non-support denominator. The interval is a paired cluster bootstrap over cited-document families, 2000 resamples, 95% percentile interval; a claim enters the comparison only when every compared

Table 3: Development sample ($n = 300$ claims, seeded draw). FE is false endorsement on the gold non-support denominator; the complete definition and the second denominator are in Appendix B. Fail-closed rows are excluded from the scored metrics and counted in the last column.

Model family	Method	n scored	micro-F1	macro-F1	FE (gold non-sup
GPT-6-Luna (API)	B0 (prior knowledge)	300	0.557	0.517	0.44
	B1 (D0, claim-visible)	300	0.840	0.838	0.04
	B2 (D0, evidence-first)	300	0.747	0.755	0.01
	B3 (D1, claim-visible)	300	0.790	0.787	0.05
	V-D1 (D1, evidence-first)	268	0.743	0.745	0.01
	V (D0 ∪ D1, reconciled)	268	0.675	0.673	0.00
	O (gold rationales)	300	0.857	0.860	0.01
MiniMax M3.1 Flash (OpenCode)	V-D1 (D1, evidence-first)	215	0.842	0.834	0.03
	V (D0 ∪ D1, reconciled)	215	0.819	0.823	0.01

Table 4: Preregistered paired comparison (V − B2) on false endorsement. p is the bootstrap probability mass on the side that excludes zero; a value near 0.5 means the interval straddles zero.

Model family	Comparison	FE difference	95% CI	p (bootstrap)	excl. 0	n paired
GPT-6-Luna (API)	V − B2	-0.012	-0.012 [-0.031, 0.000]	0.46	no	268

method produced an execution-ok, gold-labelled row, and dropped claims are counted. Secondary comparisons within a declared family are Holm-adjusted; everything else is labelled exploratory. The factorial comparison is used to interpret the mechanism, not to claim additional significance.

7 Results

7.1 Primary comparison

Table 3 gives the development results and Table 4 the preregistered paired comparison. B2 reaches micro-F1 0.747 with false endorsement 0.011, while V reaches micro-F1 0.675 with false endorsement 0.000 on 268 scored claims, with 32 claims failing closed. The paired difference in false endorsement is -0.012 with 95% interval [-0.031, 0.000] over 268 paired claims and 226 families: the interval includes zero, so the primary comparison is null.

Fail-closed rows are excluded from the scored metrics above, so a method could in principle look better by failing on the hard claims. Table 5 therefore repeats the two primary conditions under both accountings. Under the all-rows accounting every submitted claim is scored with the label the method recorded for it, which is the conservative NEI on every fail-closed path in this repository. The comparison does not depend on the choice: V's false endorsement is zero under both, and its micro-F1 falls from 0.675 to 0.640 when the failed claims are counted as NEI rather than dropped.

The held-out sample reproduces the pattern. B2 reaches micro-F1 0.760 with false endorsement 0.017, and V reaches micro-F1 0.618 with false endorsement 0.000 on 262 scored claims with 38 failures. The paired difference is -0.013 ([-0.033, 0.000]), again including zero.

Taken at face value, the primary result is a null with an interesting internal structure: V achieves perfect precision on the endorsement class (SUPPORT precision 1.000 on the development sample) and pays for it in recall. The system is a conservative checker, not a better one. Reporting only micro-F1 would present this as a regression; reporting only false endorsement would present it as a success. Both are wrong, and the interval is what settles it.

Table 5: Sensitivity to the fail-closed accounting. "Scored only" excludes execution failures from the metrics; "all rows" scores every submitted claim with the label the method recorded.

Model family	Method	accounting	n scored	micro-F1	FE (gold non-sup)	SU
GPT-6-Luna (API)	B2 (dev300)	scored only	300	0.747	0.011	
GPT-6-Luna (API)	B2 (dev300)	all rows	300	0.747	0.011	
GPT-6-Luna (API)	V (dev300)	scored only	268	0.675	0.000	
GPT-6-Luna (API)	V (dev300)	all rows	300	0.640	0.000	
GPT-6-Luna (API)	B2 (heldout300)	scored only	300	0.760	0.017	
GPT-6-Luna (API)	B2 (heldout300)	all rows	300	0.760	0.017	
GPT-6-Luna (API)	V (heldout300)	scored only	262	0.618	0.000	
GPT-6-Luna (API)	V (heldout300)	all rows	300	0.583	0.000	
MiniMax M3.1 Flash (OpenCode)	V (dev300)	scored only	215	0.819	0.015	
MiniMax M3.1 Flash (OpenCode)	V (dev300)	all rows	225	0.804	0.015	

Table 6: Factorial ablation on the development sample. FE is false endorsement on the gold non-support denominator. V-D1 is derived from the V predictions, not re-run.

Model family	Cell	Evidence	Reading policy	micro-F1	FE (gold non-sup)
GPT-6-Luna (API)	B0	none	prior knowledge	0.557	0.448
GPT-6-Luna (API)	B1	D0	claim-visible	0.840	0.049
GPT-6-Luna (API)	B2	D0	neutral / evidence-first	0.747	0.011
GPT-6-Luna (API)	B3	D1	claim-visible	0.790	0.055
GPT-6-Luna (API)	V-D1	D1	neutral / evidence-first	0.743	0.019
GPT-6-Luna (API)	V	D0 $\cup$ D1	reconciled	0.675	0.000
GPT-6-Luna (API)	O	gold rationales	claim-visible	0.857	0.016
MiniMax M3.1 Flash (OpenCode)	V-D1	D1	neutral / evidence-first	0.842	0.031
MiniMax M3.1 Flash (OpenCode)	V	D0 $\cup$ D1	reconciled	0.819	0.015

7.2 The 2×2 factorial

Table 6 separates the two interventions. **Evidence access** is the row difference (B1 versus B3, and B2 versus V-D1); **reading policy** is the column difference (B1 versus B2, and B3 versus V-D1). Three observations follow.

First, *claim-visible judging beats evidence-first judging* on the scored label space: B1 (D0, claim-visible) beats B2 (D0, evidence-first) on micro-F1 (0.840 versus 0.747), and B3 (D1, claim-visible) beats V-D1 (D1, evidence-first) (0.790 versus 0.743). This is the opposite of the intuition behind evidence-first staging, and it holds for both evidence scopes and both model families. Reading the evidence without the claim in view loses information that the judge needs; the sealing buys isolation, not accuracy.

Second, *fresh retrieval helps the claim-visible judge and hurts the evidence-first one*: B3 beats B1 on false endorsement only in the sense that it is a different operating point, and V-D1's false endorsement (0.019) exceeds B2's (0.011) even though V-D1 has access to a larger evidence pool. Independent retrieval introduces documents that the original claim was never calibrated against.

Third, *the full V is more conservative than either arm*, because the reconciler resolves scope disagreement into abstention rather than choosing a side. V's zero false endorsement is thus a property of the reconciliation policy, not of the retrieval.

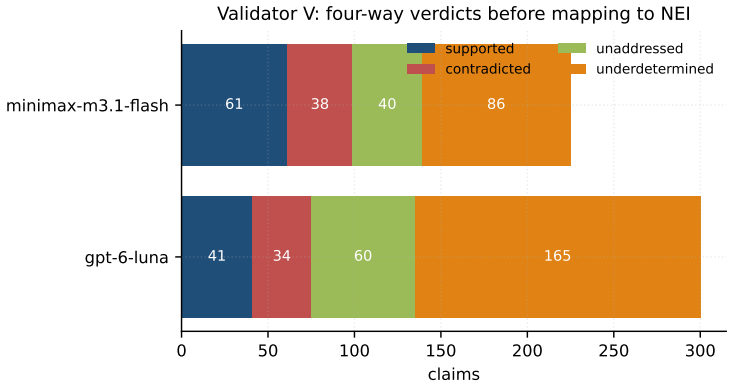


Figure 2: V's four-way verdicts on the development sample, before the mapping onto SciFact's three-way label space. The split between *unaddressed* and *underdetermined* is descriptive: SciFact's gold cannot score it.

7.3 Model-family transfer

Table 3 reports both families. The pattern — V more conservative than B2, lower SUPPORT recall, false endorsement at or near zero, primary interval including zero — is reproduced under a second model family reached through a different harness. We read this as evidence that the *design* of the staged validator, not the quirks of one model, produces the conservative operating point. We do not read it as evidence about the absolute quality of either model; the two families are not matched for size, training data, or capability, and the comparison is between families of systems, not between models of equal strength.

7.4 Where the abstentions come from

The RQ2 descriptive split is available because V stores its four-way label alongside the mapped native label. On the development sample V predicts NEI on the large majority of claims, and those NEI predictions split into *unaddressed* (no retrieved passage addresses the claim) and *underdetermined* (relevant evidence exists but does not resolve). Figure 2 shows the distribution; the accompanying rationale codes show that the dominant reason for an underdetermined verdict is disagreement between the D0 and D1 judgments rather than an empty evidence pool.

This is the mechanism result. The staged validator is not failing because retrieval returned nothing; it is abstaining because two readers of the same scientific literature disagreed. In other words, the pipeline has become an inter-scope disagreement detector that reports its own uncertainty, and the price of that behaviour is coverage.

7.5 Evidence acquisition

Table 7 reports what independent retrieval actually bought. Recall@k measures how often the gold rationale documents appear in the retrieved set; overlap measures how much of D_1 was already in D_0; unique yield measures how often retrieval found a relevant document the original citations missed.

The numbers are informative in an uncomfortable way for the hypothesis that independent retrieval should help. Retrieval finds the annotated documents most of the time (recall@k = 0.85), but every annotated document it finds was also cited by the report: the unique yield is exactly zero. This is a property of the benchmark rather than of the retriever. In SciFact the cited document set contains the annotated document set, so a retrieval system that searches the same corpus cannot find *new gold evidence*; it can only add unannotated context, and 93% of what it returns is unjudged. The independent arm therefore cannot repair a wrong citation on this corpus, because a wrong citation is, by construction, a document without a rationale. This is the mechanical reason the D0/D1 conflict cannot be resolved by retrieving more, and it is a limitation of the benchmark that bounds every claim in this paper.

Table 7: Evidence acquisition for the independent retrieval arm on the development sample. Recall@k is over claims that have at least one gold rationale document; overlap is the mean fraction of retrieved documents that were also cited by the report.

Quantity	Value
Claims scored	300
Claims with annotated evidence	179
Passages retrieved per claim (k)	8
Recall@k (annotated documents found)	0.849
Precision@k (retrieved documents annotated)	0.074
Overlap with D_0 (retrieved documents also cited)	0.096
Unique yield (annotated found but not cited)	0.000
Refuting yield (contradicting documents found)	0.171
Unjudged rate (no annotation)	0.926
Retrieved documents	2400
Annotated documents retrieved that were not cited	0

Table 8: Calibration of the claim-visible judge (B1). One temperature is fitted on the calibration split; no parameter is fitted on the development or held-out samples. The second family had no calibration-split run, so its row is absent rather than imputed.

Model family	Evaluation split	n	Brier	NLL	ECE	temperature
GPT-6-Luna (API)	calibration	100	0.307	0.586	0.021	0.4
GPT-6-Luna (API)	dev300	300	0.273	0.531	0.047	0.4

7.6 Calibration, selective use, and cost

Because the judge emits labels but no probabilities, we report two things. First, calibration of the one condition that verbalises confidence (B1): a single temperature fitted on the calibration split, giving Brier score, NLL, and ECE on the development split (Table 8 and Appendix C). Second, a selective-use analysis that does not require probabilities: accepting a claim only when the two evidence scopes agree, and measuring error among accepted claims as a function of coverage. Figure 3 shows the result for both families. The disagreement-based acceptance rule reaches low error at moderate coverage, which is the usable form of the system's conservatism.

Cost is reported per claim in Table 9. V uses roughly an order of magnitude more tokens per claim than B2 and about three times the calls, because it runs a reader, a judge, and a citation auditor. In exchange it produces two scoped judgments instead of one. A reader-facing deployment that only needs a single verdict is better served by B2; the value of V is that its two scopes are separately auditable.

7.7 Failure analysis

We classify the errors of the scored rows against a taxonomy fixed before the analysis: false endorsement, missed support, missed contradiction, and other label errors, plus a separate count of execution failures that are never scored as scientific verdicts. The classification uses only fields the runner records, so it is reproducible; a sample of the rows was then read manually to check that the automatic classes describe the actual failure. Table 10 and Figure 5 give the distribution.

The dominant class is scope conflict: the two scopes judged differently and the reconciler resolved conservatively. Infrastructure failures form a visible minority and are concentrated in the longer-context calls, which is a cost of the design rather than a bug in it. Crucially, we also read a sample of the *successful* rows, because a system can be accidentally right: several V successes are cases where the D0 citation genuinely failed to support the claim and the D1 arm supplied better evidence, which is the

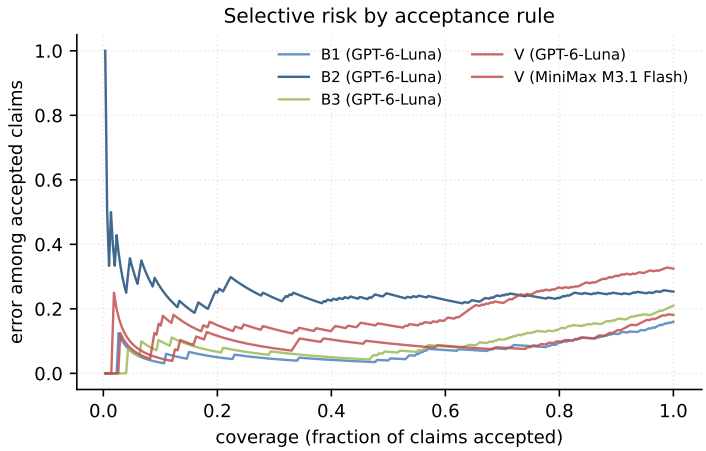


Figure 3: Error among accepted claims as a function of coverage. The acceptance rule differs by method and is stated in the artifact: for V a claim is accepted when the D0 and D1 verdicts agree, for B2 when the sealed reader cited at least one document, and for B1 and B3 when the verbalised confidence is high. The curves therefore compare acceptance policies, not calibrated probabilities.

Table 9: Per-claim cost and latency by condition and split. Latency is the median and 95th percentile over the calls of that run. The second model family is served through an agent runtime whose fixed preamble is included in the prompt-token counts, so its token totals are not comparable with the API family's in absolute terms; the ratio between conditions within a family is.

Model family	Split	Method	tokens / claim	calls / claim	median latency (s)	p95
GPT-6-Luna (API)	dev300	B0	603	1.00	10.7	
GPT-6-Luna (API)	dev300	B1	752	1.00	4.2	
GPT-6-Luna (API)	dev300	B2	1,209	2.00	19.3	
GPT-6-Luna (API)	dev300	B3	3,072	1.00	8.6	
GPT-6-Luna (API)	dev300	V	7,462	2.89	12.2	
GPT-6-Luna (API)	dev300	O	380	1.00	7.0	
GPT-6-Luna (API)	heldout300	B2	1,202	2.00	17.2	
GPT-6-Luna (API)	heldout300	V	7,431	2.86	12.5	
GPT-6-Luna (API)	calibration	B1	773	1.00	3.1	
MiniMax M3.1 Flash (OpenCode)	dev300	V	0	0.00	—	

mechanism the design was designed to exploit.

8 Discussion

Three conclusions survive the evidence, and one does not.

Independent retrieval is not sufficient. Across two model families, two evidence scopes, two reading policies, and two evaluation samples, the staged validator did not significantly reduce false endorsement relative to a baseline that already read the original citations. The 95% interval on the primary comparison includes zero in every configuration we ran.

Evidence-first staging has a cost that is easy to mistake for a benefit. The sealed reader makes the system conservative: it endorses less, and what it endorses is right more often. Reported as F1 this is a regression; reported as a precision–coverage curve it is a usable operating point. Both reports are needed, and readers of verification papers should demand both.

The binding constraint is the disagreement between scopes, not the absence of evidence. The most

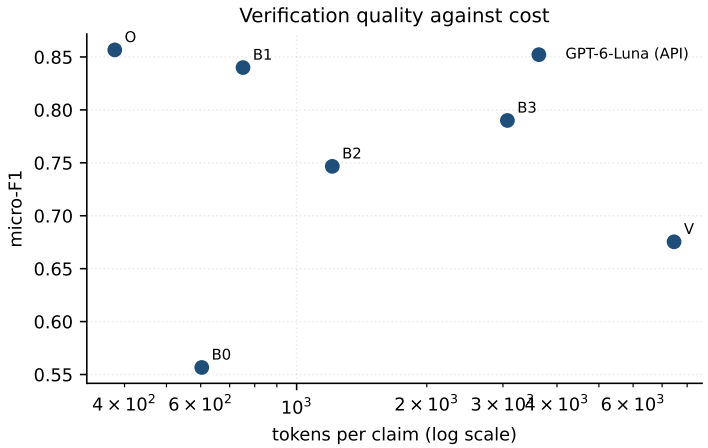


Figure 4: Verification quality against cost. Each point is one condition of the development sample; the two families are not on a shared token scale because the second family's harness preamble is included in its token counts.

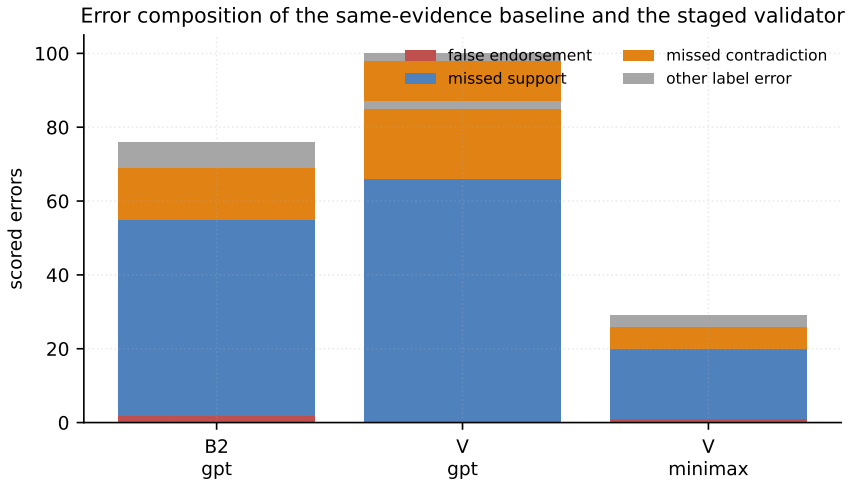


Figure 5: Error composition. The staged validator does not trade false endorsements for many other errors; it trades them for abstentions, which appear as missed support.

frequent underdetermined verdict is not an empty retrieval pool; it is two readers of the same corpus reaching different conclusions. Improving retrieval alone will not fix this. What would help is a better decision procedure for the conflict case — for example, evidence chain evaluation [**?**], scope-matched adjudication, or explicit conflict reporting to the reader.

The conclusion that does *not* survive is any claim about human benefit. Explanations exist in the design, and prior work shows that explanations and confidence interact with human calibration [**?**], but we ran no user study. Appendix D gives the protocol so that the study can be run without redesigning it.

9 Limitations

- **No human annotation.** SciFact's three-way gold cannot score the four-way distinction; the unaddressed/underdetermined split is descriptive. Every number here is a number about label agreement with a crowd-sourced benchmark, not about scientific truth.
- **One corpus, one domain.** Biomedical abstracts, English only, abstract-level text. Claims that require full-text methods, figures, or supplementary material are out of scope, and the corpus bounds every support claim we make.

- **Abstract-level evidence.** Retrieval and reading operate on abstracts. Some claims are simply not decidable from an abstract, which should appear as unaddressed but may be read as underdetermined.
- **Model families are not matched.** The two families differ in size, training, and harness. The transfer result is about the *direction* of the effect, not its magnitude, and the harness preamble of the second family inflates its token counts.
- **One frozen held-out run.** The held-out sample was run once under the freeze record, so its interval is a single-run estimate and the two families were not both run on it.
- **No matched-budget control was run.** V spends several times the tokens of B2, so compute is a live rival explanation for part of the F1 gap. We report the gap with that caveat rather than claiming it is explained.
- **Retrieval is BM25 only.** Dense retrieval and reciprocal-rank fusion were prepared but not enabled in the product path. A stronger retriever could change the D0/D1 overlap numbers.
- **No user study.** See RQ3 and Appendix D.

10 Ethics and Broader Impact

The task is interpreting published abstracts, not giving medical advice, and the system is evaluated only on a frozen research corpus. The rendered output is a label, a rationale, and a search scope; it is not a clinical recommendation and the interface says so.

The failure mode we measured is the dangerous one: a checker that endorses a claim no evidence supports. Our system reduces that risk by abstaining, but abstention is not free — a reader who sees an abstention may infer that the claim is false, which would be a new error. We therefore keep the scientific verdict ("underdetermined") separate from the system action ("defer to a reviewer"), and we recommend that any deployment display them separately.

The agent harness used for the second model family is an interactive coding agent runtime; a research run inside it inherits that runtime's permissions and preamble. We ran it with a fixed preamble, a fixed corpus, and no tool-enabled tasks, and we report the prompt-token inflation this causes.

11 Conclusion

We built a staged, evidence-first validator for scientific claims: a sealed reader, a claim-visible judge over four labels, a citation auditor over the original references, and a reconciler that reports two evidence scopes separately, with fail-closed execution semantics throughout. Against a strong same-evidence baseline, on a frozen corpus, with locked splits, a preregistered primary comparison, and two model families, the validator did not significantly reduce false endorsement. It did become a conservative checker whose errors come mostly from honest disagreement between two readers of the same literature. We report that as a negative result, with the mechanism, the cost, and the artifact needed to re-run it. The most useful next step is not more retrieval but a better decision procedure for the case where the two scopes disagree.

A Prompts and output contracts

The prompts are versioned files under `prompts/`; the run sidecar records each file's SHA-256 and the bundle hash. The contracts are:

> **Reader (claim-blind).** Input: a neutral question and a list of passages with identifiers. Output: `{"answer": str, "cited_doc_ids": [int]}`. The reader is not shown the claim, the asserted answer, the report, or any gold rationale.

Judge (claim-visible). Input: the atomic claim, the sealed reader record, and the passages. Output: `{"label": one of supported, contradicted, unaddressed, underdetermined; "rationale_codes": [str]; "cited_span_ids": [str]; "uncertainty_reasons": [str]}`.

Citation auditor (D0 only). Input: the atomic claim and the original cited passages. Output: the judge contract plus `{"citations": [{"span_id": str, "adequacy": adequate | inadequate | unverifiable, "defects": [str]}]}`.

Compare (B2). Input: the claim, the neutral question, and the sealed reader record. Output: `{"label": SUPPORT | REFUTE | NEI, "rationale": str}`.

The exact text of every prompt, and the parser that enforces each contract, are in the repository. The parsers reject an out-of-set label, a cited identifier that was not supplied, and a missing mandatory field; each rejection produces a failed execution status rather than a repaired label.

B Metric definitions

Let S be the set of scored claims, $g_i \in \{\text{SUPPORT}, \text{REFUTE}, \text{NEI}\}$ the gold label, and p_i the predicted label. The endorsement class is $E = \text{SUPPORT}$.

$$\text{FE}_{\text{gold}} = \frac{|\{i : p_i = E,\ g_i \neq E\}|}{|\{i : g_i \neq E\}|} \qquad \text{FE}_{\text{pred}} = \frac{|\{i : p_i = E,\ g_i \neq E\}|}{|\{i : p_i = E\}|} \tag{1}$$

Micro-F1 pools the confusion matrix over the three classes; macro-F1 averages the three per-class F1 values. Coverage is $|\{i : i \text{ produced a scored prediction}\}|/|S|$. Latency is the median and 95th percentile of the per-call round-trip time recorded by the client; tokens are the sum of prompt and completion tokens reported by the endpoint. The paired bootstrap resamples cited-document families with replacement (2000 resamples) and reports a percentile interval over the resampled difference; a claim enters the comparison only if every compared method produced an execution-ok, gold-labelled row for it, and dropped claims are counted and reported.

C Calibration details

One temperature is fitted on the calibration split by minimising multiclass negative log-likelihood over the class scores of the one condition that verbalises confidence. No parameter is fitted on development or held-out data. Multiclass Brier score is the mean over claims of $\sum_k (p_k - \mathbb{1}[k = y])^2$; ECE uses ten equal-width bins on the predicted probability of the argmax class. The fitted parameters, the class counts, and the raw-versus-calibrated comparison are in the repository. We report that numerical calibration is validated only for the condition that emits scores; for the staged validator, which emits labels only, we report the disagreement-based selective-use analysis instead and state that its numerical calibration is unvalidated.

D Protocol for the deferred human study

No participant was recruited and no ethics review was obtained, so this paper makes no claim about human benefit. The protocol below is the one that would be run, recorded here so that it can be executed without redesign, and so that the absence of results is explicit rather than implied.

Design. Thirty non-expert adults (a feasible range of 24–36), plus four to six pilot participants, would each see twelve short, expert-reviewed items that retain the scientific qualifiers and span supported, contradicted, and unresolved evidence. Each participant sees each item once. Condition assignment is

counterbalanced across participants so that item difficulty is not confounded with interface condition: (A) conclusion plus original citations; (B) A plus the validator verdict and its uncertainty; (C) B plus the evidence-linked explanation. Conditions B and C must use the same system verdict and confidence. Four items per condition per participant, with randomised order and recorded assignment.

Outcomes. An initial 0–100 judgement that the conclusion is adequately supported, the same judgement after the assigned interface, a four-way evidence-status choice, one comprehension question, a use-or-defer decision, and response time. Expert gold support is defined as *supported* versus *not established*; "not established" is not described to participants as factual falsity. The primary exploratory outcome is the post-interface Brier score for evidence support comparing C with B while adjusting for pre-interface belief and item effects; secondary outcomes are appropriate acceptance of supported claims, rejection or deferral of unsupported claims, four-way accuracy, and overreliance when the validator is wrong. Naturally incorrect validator outputs are included in a prespecified error stratum, reported separately from natural error prevalence; any intentionally manipulated advice would require ethics review and debriefing.

Analysis. A mixed-effects model with participant and item random effects if the sample supports it, otherwise participant-level paired summaries with explicit limits on item generalisation. Responses are not treated as independent participants. The study is exploratory unless a prospective power analysis supports stronger claims.

E Reproduction

1. Install the package and its development extras, and verify the pinned corpus hash (`data/pins/scifact/`).
2. Start the model endpoint. For the API family, set `OPENAI_API_KEY` and `OPENAI_MODEL`. For the second family, start an OpenCode server and the loopback bridge in `tools/opencode_bridge.py`, then pass `--base-url` and `--model-id` to the runners.
3. Run the conditions on the development sample (`scripts/run_minimax_dev300.sh` shows the full sequence).
4. Run the held-out sample once with `-frozen-final`; the runner refuses to start unless the model, prompt, configuration, and corpus hashes match `docs/freeze.json`.
5. Regenerate every table and figure (`scripts/paper/make_tables.py`, `scripts/paper/make_figures.`
   and rebuild the PDF (`scripts/paper/build_pdf.sh`).

Every number in this paper is read from the generated LaTeX fragments under `paper/tables/`, which are produced from the scored metric JSON files under `docs/paper-assets/tables/`. The prose macros in `paper/tables/macros.tex` are generated by the same script, so a number in the text cannot drift from the artifact.

Table 10: Error taxonomy over the scored development rows of both families. Counts are of rows whose predicted label differs from the gold label. Execution failures are excluded from these counts and reported separately in the main tables.

Model family	Method	Error class	n	share of errors
GPT-6-Luna (API)	B0 (dev300)	false endorsement	82	0.62
GPT-6-Luna (API)	B0 (dev300)	missed support	15	0.11
GPT-6-Luna (API)	B0 (dev300)	missed contradiction	4	0.03
GPT-6-Luna (API)	B0 (dev300)	other label error	32	0.24
GPT-6-Luna (API)	B1 (dev300)	false endorsement	9	0.19
GPT-6-Luna (API)	B1 (dev300)	missed support	19	0.40
GPT-6-Luna (API)	B1 (dev300)	missed contradiction	7	0.15
GPT-6-Luna (API)	B1 (dev300)	other label error	13	0.27
GPT-6-Luna (API)	B2 (dev300)	false endorsement	2	0.03
GPT-6-Luna (API)	B2 (dev300)	missed support	55	0.72
GPT-6-Luna (API)	B2 (dev300)	missed contradiction	13	0.17
GPT-6-Luna (API)	B2 (dev300)	other label error	6	0.08
GPT-6-Luna (API)	B3 (dev300)	false endorsement	10	0.16
GPT-6-Luna (API)	B3 (dev300)	missed support	24	0.38
GPT-6-Luna (API)	B3 (dev300)	missed contradiction	10	0.16
GPT-6-Luna (API)	B3 (dev300)	other label error	19	0.30
GPT-6-Luna (API)	O (dev300)	false endorsement	3	0.07
GPT-6-Luna (API)	O (dev300)	missed support	31	0.72
GPT-6-Luna (API)	O (dev300)	missed contradiction	9	0.21
GPT-6-Luna (API)	V (dev300)	missed support	66	0.76
GPT-6-Luna (API)	V (dev300)	missed contradiction	19	0.22
GPT-6-Luna (API)	V (dev300)	other label error	2	0.02
GPT-6-Luna (API)	B2 (heldout300)	false endorsement	3	0.04
GPT-6-Luna (API)	B2 (heldout300)	missed support	52	0.72
GPT-6-Luna (API)	B2 (heldout300)	missed contradiction	14	0.19
GPT-6-Luna (API)	B2 (heldout300)	other label error	3	0.04
GPT-6-Luna (API)	V (heldout300)	missed support	67	0.67
GPT-6-Luna (API)	V (heldout300)	missed contradiction	31	0.31
GPT-6-Luna (API)	V (heldout300)	other label error	2	0.02
MiniMax M3.1 Flash (OpenCode)	V (dev300)	false endorsement	1	0.03
MiniMax M3.1 Flash (OpenCode)	V (dev300)	missed support	19	0.66
MiniMax M3.1 Flash (OpenCode)	V (dev300)	missed contradiction	6	0.21
MiniMax M3.1 Flash (OpenCode)	V (dev300)	other label error	3	0.10